llmXive Automated Review of Edit-Compass & EditReward-Compass: A Unified Benchmark for Image Editing and Reward Modeling

llmXive automated review panel

Please read first — this review is fully automated and not checked by any human. It is written by free, open-weight LLM agents that read the paper once, so **errors, misreadings, and inaccuracies are likely**. Nothing here has been verified by a person. Treat every point below as a fast, fallible first-pass signal — not an authoritative assessment.

How this review was produced

This Reviewed Preprint was read by a panel of specialist llmXive LLM reviewers. Each reviewer is deliberately confined to a *single narrow lens* — writing, logic, statistics, and so on — and does not comment outside it, so that distinct concerns are surfaced independently rather than blurred into one overall score. The panel runs *once* over the paper. Every reviewer's exact instructions are public in the llmXive agent-prompt library; each section below also links the specific prompt it used.

This report is **advisory, automated feedback for the authors**. llmXive does not modify the paper, and **nothing is accepted or rejected** on its basis — every point below is a suggestion the authors are free to weigh. Each reviewer's section also names the underlying model that produced it.

This paper's panel (9 lenses): Claim Accuracy, Figure Critic, Jargon Police, Logical Consistency, Overreach, Safety Ethics, Scientific Evidence, Statistical Analysis, and Writing Quality. Each lens is described in its own section below, alongside the model and the exact prompt it used.

Claim Accuracy

Verifies factual claims against their citations — whether each cited source actually supports the claim attributed to it, and whether any claim is stated more strongly than its evidence permits.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_claim_accuracy.md`

The paper presents a comprehensive benchmark, but several factual claims and citations require verification to ensure accuracy.

First, the claim in the Abstract and Introduction that "Qwen-Image-Edit" scores 2.69 is inconsistent with Table 1 (tab:Image Editing Bench Main Results_EN), which lists "Qwen-Image-Edit-2511" with a score of 2.69. The text does not specify the version "2511," which could mislead readers about the model's identity. The authors should clarify whether "Qwen-Image-Edit" and "Qwen-Image-Edit-2511" are the same model or if the text requires a specific version citation.

Second, the claim in Section 5.2 that "Thinking-enabled inference improves scores significantly (e.g., +9.83 for Qwen3.5-9B)" is not supported by the data in Table 1 (tab:RMBench Main Result). The table shows Qwen3.5-9B (0.6016) vs. Qwen3.5-9B^‡ (0.6681), a difference of 0.0665 (6.65 percentage points), not 9.83. The value 9.83 appears in the "Compute Resources" section or as a raw score in a different context, but not as a percentage point gain in the main results table. This numeric claim is factually unsupported by the provided data and should be corrected or removed.

Third, the citation "nanobananapro" is used to refer to "Nano-Banana Pro" in the Abstract, Introduction, and Tables. However, the bibliography entry "nanobananapro" is a Google blog post from 2025, while the model "Nano-Banana 2" is cited as "gemini3pro" (Google DeepMind, 2025). The text conflates "Nano-Banana Pro" and "Nano-Banana 2" in some places (e.g., Table 1 lists "Nano Banana 2" and "Nano Banana Pro" as separate rows, but the text often uses them interchangeably). The authors should clarify if these are distinct models and ensure citations match the specific model version being evaluated.

Finally, the claim that "Native multimodal LLMs outperform existing reward models" (Abstract) is supported by Table 1, but the specific claim that "Qwen3.5-9B rivals larger models" lacks a clear baseline in the text. The table shows Qwen3.5-9B (0.6016) vs. Qwen3.6-27B (0.7183), which is a significant gap. The claim "rivals" is ambiguous without a specific metric or context (e.g., "rivals in efficiency" or "rivals in specific sub-tasks"). The authors should provide a more precise claim or supporting data.

These issues, while not fatal to the paper's overall contribution, require correction to ensure the accuracy of the claims and the integrity of the citations.

Action items.

- **[writing]** The claim that 'Qwen-Image-Edit' scores 2.69 (Intro) contradicts Table 1 which lists 'Qwen-Image-Edit-2511'. Verify if these are the same model or update the text to specify the version to support the claim accurately.
- **[science]** The claim that thinking-enabled inference improves scores by '+9.83' for Qwen3.5-

9B (Sec 5.2) is unsupported by Table 1, which shows a 0.0665 difference. The value 9.83 appears elsewhere but not as a gain here. Correct the number or remove the claim.

- [**writing**] The text conflates ‘Nano-Banana Pro’ and ‘Nano-Banana 2’ while citing them differently. Clarify if they are distinct models and ensure citations match the specific model version evaluated in the tables.
- [**writing**] The claim that ‘Qwen3.5-9B rivals larger models’ is ambiguous given the 0.1167 gap to Qwen3.6-27B in Table 1. Specify the context (e.g., efficiency) or provide data supporting the ‘rivals’ assertion.

Figure Critic

Inspects each figure directly — the rendered image alongside its caption — for clarity, axis labels and units, color choices, legibility at print scale, and whether the figure earns its place. This lens runs on a vision-capable model.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_figure_critic.md`

Figure 1

The figure provides a clear visual overview of the benchmark’s task categories, but the caption’s claim of ‘36 diverse tasks’ contradicts the visible 12 categories, and the inclusion of non-English text in the ‘Chemical’ example without translation reduces accessibility.

Figure 2

The figure provides a clear visual overview of the data construction pipelines, but the caption contains a grammatical error (‘in .’) and lacks specific mapping between the listed task categories and the figure’s sub-panels.

Figure 3

Figure 3 presents clear bar charts with data labels, but panel (b) suffers from ambiguous x-axis labeling that conflicts with the caption’s description of ‘evaluation protocols,’ and the y-axis label in panel (a) is poorly formatted.

Figure 4

The figure presents a qualitative comparison of subject addition, but the displayed prompt (‘Add an Adidas logo...’) contradicts the visual content (a basketball jersey). Additionally, the input image shows a blank canvas rather than the original scene, which may misrepresent the baseline for the addition task.

Figure 5

Figure 5 presents a qualitative comparison of subject removal but lacks a clear ground truth panel showing the correct result (removing only the non-smiling ball). Additionally, the task instruction is embedded as image text rather than being clearly defined in the caption or layout.

Figure 6

The figure effectively displays qualitative comparisons for the Subject Replace task, but the top instruction text contradicts the input image content (referencing a spoon instead of the visible bowl), and the label 'Bagel' appears to be a labeling error.

Figure 7

The figure presents a qualitative comparison for subject extraction but contains a factual error in the prompt (pigeon vs. goose) and includes a missing result for the OmniGen2 model, rendering the comparison incomplete.

Figure 8

The figure effectively demonstrates the 'Change Color' task, but the OmniGen2 result is a clear failure case where the model alters the entire scene rather than the specific target object. The layout is readable, though the prompt placement is unconventional.

Figure 9

The figure effectively demonstrates the 'Change Size' task across multiple models, but the 'Bagel' result shows a significant failure (object removal) that is not highlighted, and the instruction overlay obscures the top of the input panels.

Figure 10

The figure effectively displays qualitative results for the 'Change Material' task with clear model labels, but lacks an internal caption and relies on a brief prompt that does not specify the target material's visual properties.

Figure 11

The figure attempts to show qualitative comparisons for visual text editing but fails because the input panel displays the text prompt instead of the source image. Additionally, the prompt overlay is visually intrusive, and at least one model (Bagel) fails to follow the spatial constraints of the instruction.

Figure 12

The figure effectively demonstrates Chinese text editing capabilities across various models, but the inclusion of the raw instruction prompt in the top row is unprofessional and the task definition (add vs. replace) is slightly ambiguous regarding the existing English text.

Action items.

- [**writing**] Figure 1: The caption claims the figure covers 36 diverse tasks, but the visual content displays 12 distinct task categories (e.g., 'Subject Addition', 'Action', 'Knapsack'). The discrepancy between the stated count (36) and the visible categories (12) is confusing and likely inaccurate.
- [**writing**] Figure 1: The 'World Knowledge Reasoning' section contains a 'Chemical' task with a blackboard image containing Chinese text, but the caption does not specify the language or provide a translation, making the specific task instruction illegible to non-Chinese speakers.
- [**writing**] Figure 2 caption contains a grammatical error: 'pipelines in .' includes a dangling preposition with no object.
- [**writing**] Figure 2 caption is vague; it lists task categories (e.g., 'Dynamic Manipulation') but does not explicitly map them to the specific sub-panels (a, b, c) shown in the figure.
- [**science**] Figure 3(b): The x-axis labels ('ImgEdit-Bench', 'GEdit-Bench', 'RISE-Bench', 'Ours') are inconsistent with the caption's claim of 'evaluation protocols'; the labels imply specific datasets rather than protocols, creating ambiguity about what is being compared.
- [**writing**] Figure 3: The y-axis label in panel (a) ('Pearson correlation coefficient') is rotated 90 degrees and difficult to read; consider horizontal alignment or better spacing.
- [**science**] Figure 4: The prompt 'Add an Adidas logo to the side of the white truck box' does not match the visual content, which shows a framed basketball jersey being added to a wall. The caption 'Subject Addition' is generic, but the specific prompt displayed is factually inconsistent with the image shown.
- [**science**] Figure 4: The 'Input Image' shows a blank white canvas on the wall, yet the task is 'Subject Addition'. A more appropriate baseline for this task would be the original scene with the empty wall (no canvas) to demonstrate the addition of the object itself, rather than the addition of a canvas to a blank space.
- [**science**] Figure 5: The 'Input Image' contains four balls (one sad, three smiling), but the instruction 'Remove the balls... that are not smiling' implies removing only the single sad ball. The ground truth should therefore show three smiling balls. However, models like 'Nano Banana 2' and 'SeeDream4.5' are shown removing the smiling balls and keeping the sad one, or removing all but one smiling ball, which contradicts the prompt's logic. The figure fails to provide a clear 'Ground Truth' panel showing th
- [**writing**] Figure 5: The instruction text 'Remove the balls in the image that are not smiling' is embedded directly into the image layout rather than being described in the caption or a separate text box, which is inconsistent with standard scientific figure presentation.

- **[writing]** Figure 6: The top instruction text ('Replace the spoon with a Dove chocolate.') is inconsistent with the visual evidence; the input image contains a bowl of strawberries, not a spoon, making the instruction confusing or incorrect for the displayed task.
- **[writing]** Figure 6: The label 'Bagel' for the first result column appears to be a typo or artifact, as 'Bagel' is not a known image editing model and does not fit the naming convention of the other models (e.g., OmniGen2, Flux2-Dev).
- **[science]** Figure 7: The task instruction explicitly requests to 'Extract the larger pigeon', yet the input image contains two geese (not pigeons). This mismatch between the prompt and the visual data undermines the validity of the qualitative comparison.
- **[science]** Figure 7: The 'OmniGen2' result panel is empty (blank white), failing to provide a visual result for comparison, which makes it impossible to evaluate the model's performance on this specific task.
- **[writing]** Figure 7: The instruction box at the top is not part of the standard figure layout (unlike the input image or model outputs) and appears to be a raw prompt overlay rather than a formal figure element.
- **[science]** Figure 8: The 'OmniGen2' result fails the task by changing the color of all objects (bottles, jars) to blue, not just the toothbrush handle as requested in the prompt.
- **[writing]** Figure 8: The instruction prompt is displayed in a large banner at the top rather than being associated with the specific input image, which is less standard for qualitative comparison figures.
- **[science]** Figure 9: The 'Bagel' result fails the prompt by scaling the robot down but also removing the orange robot entirely, whereas other models preserve both subjects. This makes the comparison unfair without noting the failure mode.
- **[writing]** Figure 9: The instruction box at the top is not part of the original input image but is overlaid on the entire figure, obscuring the top edge of the 'Input Image' and 'Bagel' panels.
- **[writing]** Figure 10: The figure contains no caption text within the rendered image itself; the provided caption 'Figure 10: Qualitative comparisons on the Change Material task' is external metadata. The image relies entirely on the prompt 'Make the side table ceramic.' at the top, which is not a formal caption.
- **[science]** Figure 10: The prompt 'Make the side table ceramic' is ambiguous regarding the specific material properties (e.g., glossy vs. matte, white vs. colored ceramic) to be applied, making it difficult to objectively judge if models like 'Bagel' or 'OmniGen2' failed or succeeded in capturing the intended material.
- **[science]** Figure 11: The 'Input Image' panel displays a prompt instruction ('Add the English title...') rather than the actual source image to be edited, making it impossible to evaluate the models' performance on the visual task.

- **[science]** Figure 11: The 'Bagel' model output fails to follow the instruction to place the text on the 'wooden table', instead placing it in the empty space above the table, indicating a failure in spatial reasoning.
- **[writing]** Figure 11: The instruction prompt is rendered as a large, distracting overlay on the input panel, obscuring the visual context and reducing the professional quality of the figure.
- **[science]** Figure 12: The top row contains a dashed instruction box ('Add large white handwritten text...') that is not an image result but a prompt artifact; this should be removed or clearly separated from the model outputs to avoid confusion.
- **[writing]** Figure 12: The 'Input Image' panel contains the English text 'So Young' which is not addressed in the Chinese editing instruction, yet the Chinese text is added to the sky; the caption should clarify if the task is 'add text' or 'replace text'.

Jargon Police

Flags needless jargon: terms with plainer equivalents, acronyms not defined at first use, and passages that needlessly exclude non-specialist readers.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_jargon_police.md`

The manuscript relies heavily on domain-specific acronyms and jargon that are not defined at their first occurrence, creating a barrier for readers outside the immediate sub-field of image editing benchmarks.

In **Table 1** (Section 1), the abbreviations **AVR** (Algorithm Visual Reasoning), **MIA** (Multi-Image Awareness), **WKR** (World Knowledge Reasoning), **DM** (Dynamic Manipulation), and **HP** (Human Preference) are used in the column headers and the caption without prior definition in the main text. While the caption attempts to define them, the Introduction text also uses these acronyms (e.g., "Algorithmic Visual Reasoning Tasks") without establishing the shorthand first. This forces the reader to cross-reference the table constantly.

In **Section 5 (Experiments)**, the term **FlowGRPO-inspired** is used to describe the sampling strategy. "FlowGRPO" is a specific algorithmic term (likely referring to Flow Matching + Group Relative Policy Optimization) that is not defined in the paper. The text assumes the reader knows this specific combination, which is not standard general knowledge.

The phrase **MLLM-as-judge** appears in the **Evaluation Pipeline** section. While "MLLM" is a common acronym, the compound term "MLLM-as-judge" is a specific methodological shorthand that should be expanded to "using Multimodal Large Language Models as automated judges" upon first use to ensure clarity for a broader audience.

Additionally, **rubric-based evaluation** in the Abstract and **chain-of-thought reasoning** in the Introduction are used as technical descriptors. While "chain-of-thought" is becoming more common, explicitly linking it to "step-by-step logical reasoning" would improve accessibility. The term "rubric" is also specific to assessment; a brief parenthetical explanation (e.g., "structured scoring criteria") would be beneficial.

Finally, the **Appendix** references **T.C.R.V.** (Task, Constraints, Requirements, Verification) in the system prompts without defining the acronym in the main body text where the evaluation protocol is introduced.

To improve accessibility, the authors should define all acronyms at their first mention in the main text and expand technical methodological terms (like "FlowGRPO-inspired" or "MLLM-as-judge") into plain English descriptions.

Action items.

- [**writing**] The manuscript relies heavily on domain-specific acronyms and jargon that are not defined at their first occurrence, creating a barrier for readers outside the immediate sub-field of image editing benchmarks. In Table 1 (Section 1), the abbreviations AVR (Algorithm Visual Reasoning), MIA (Multi-Image Awareness), WKR (World Knowledge Reasoning), DM (Dynamic Manipulation), and HP (Human Preference) are used in the column headers and the caption without prior definition in the main text. While the

Logical Consistency

Checks that each conclusion follows from its premises, flagging internal contradictions and causal claims that lack a stated supporting mechanism.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_logical_consistency.md`

The manuscript demonstrates strong logical consistency throughout its argumentation and experimental design. The central claim—that existing benchmarks lack the difficulty and granularity to reflect human judgment for frontier models—is logically supported by the introduction of \bench and \rmbench, which explicitly address these gaps through fine-grained task taxonomies and preference pair construction.

The causal link between the benchmark design and the observed results is well-established. The paper argues that the inclusion of complex reasoning tasks (e.g., Algorithmic Visual Reasoning, World Knowledge) reveals specific weaknesses in current models. The experimental results in Tables 1, 2, and the supplementary tables consistently show lower scores for these specific categories across both open-source and proprietary models, validating the premise that these areas are under-evaluated by prior benchmarks.

Furthermore, the conclusion that native multimodal LLMs outperform explicit reward models follows logically from the evaluation protocol. The \rmbench design, which simulates RL scenarios with preference pairs, directly tests the models' ability to align with human preferences. The results showing Qwen3.5 and Qwen3.6 outperforming specialized reward models (EditScore) are consistent with the hypothesis that generalist multimodal reasoning capabilities are transferable and effective for reward modeling.

There are no internal contradictions detected. The distinction between the two benchmarks (\bench for generation/editing capability, \rmbench for reward modeling) is maintained consistently. The evaluation metrics (Instruction Awareness, Visual Consistency, Visual Quality) are applied uniformly across the reported experiments. The reliance on MLLM-as-judge is acknowledged as a limitation, but the paper logically mitigates this by reporting high correlation with human evaluation in the User Study (Figure \ref{User_Study}), ensuring the validity of the automated scores.

The flow from problem statement to solution (benchmark construction) to validation (experiments) is coherent and free of logical gaps. The claims regarding the performance gap between proprietary and open-source systems are directly supported by the numerical data presented in the main results tables.

Overreach

Looks for over-claiming: conclusions extrapolated beyond what the data, methods, or scope justify, and whether the paper states its limitations honestly.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_overreach.md`

The paper makes several strong claims regarding the superiority of native multimodal LLMs and the specific weaknesses of current models, but these claims often extrapolate beyond the provided evidence.

First, the abstract and Section 5.2 assert that "Native multimodal LLMs outperform existing reward models." While Table 1 supports that Qwen3.6-27B (a native MLLM) outperforms EditScore (a specialized reward model), the table also includes Gemini 3.1 Pro, which is also a native MLLM and scores higher than Qwen3.6-27B. The paper does not isolate the variable of "nativeness" from model size or proprietary status. The claim implies a structural advantage of the architecture over specialized training, but the data only shows that *some* large native models perform well. This is an over-interpretation of the results.

Second, the assertion that "Weaknesses persist in world knowledge, visual reasoning, and multi-image editing" (Abstract) is presented as a definitive finding. However, the paper presents raw scores (e.g., ~1.2-1.5 for reasoning tasks) without statistical significance testing or confidence intervals. Without error bars or a statistical test comparing these scores against a null hypothesis or a baseline variance, it is difficult to claim these are "persistent" weaknesses rather than artifacts of the specific benchmark distribution. The paper should qualify this claim or provide the necessary statistical backing.

Third, the specific claim that "Thinking-enabled inference improves scores significantly (e.g., +9.83 for Qwen3.5-9B)" (Section 5.2) is ambiguous. The value 9.83 is likely a percentage point increase, but the text does not explicitly state the baseline score or the metric (e.g., is it a 9.83% relative increase or 9.83 absolute points on a 5-point scale?). Given the scale of the scores in Table 1 (ranging from 0.4 to 0.8), a 9.83 absolute point increase is impossible, suggesting a unit confusion or missing context that renders the claim misleading.

Finally, the paper claims "High" human preference evaluation (Table 1) and "stronger agreement with human preferences" (Section 5.3) based on a human study of only 180 instances (Appendix). While this is a reasonable sample for a pilot, extrapolating this to a general claim of "stronger agreement" for the entire benchmark without reporting the correlation coefficient (r) or p-value in the main text is an overreach. The specific numbers in the User Study figure (Figure User_Study) are not visible, but the text should explicitly state the statistical strength of this correlation to justify the strong claim.

Action items.

- [**writing**] The paper makes several strong claims regarding the superiority of native multimodal LLMs and the specific weaknesses of current models, but these claims often extrapolate beyond the provided evidence. First, the abstract and Section 5.2 assert that "Native multimodal LLMs outperform existing reward models." While Table 1 supports that Qwen3.6-27B (a native MLLM) outperforms EditScore (a specialized reward model), the table also includes Gemini 3.1 Pro, which is also a native MLLM and scores hig

Safety Ethics

Examines safety and ethics — dual-use risk, IRB/IACUC and consent concerns, potential for harm, data privacy, and conflicts of interest — aiming to be thorough but not alarmist.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_safety_ethics.md`

The manuscript introduces a unified benchmark for image editing and reward modeling. From a safety and ethics perspective, the paper raises several concerns that require clarification before acceptance.

First, the **Impact Statement** (Section: supp:impact) is critically inadequate. The authors state, \"no specific societal consequences highlighted,\" which is a significant oversight given the nature of the tasks evaluated. The benchmark explicitly includes \"Emotion Change\" and \"Object Interaction\" (Section: e000, Task Taxonomy), which are high-risk categories for generating non-consensual deepfakes, harassment material, or manipulated evidence. The paper must include a dedicated discussion on the dual-use potential of these capabilities, specifically addressing how the benchmark might inadvertently facilitate the creation of harmful content and what safeguards (e.g., refusal mechanisms, watermarking) are recommended for models trained on this data.

Second, regarding **Data Privacy and Consent**, the \"Benchmark Construction\" section (Section: e000) notes the use of \"Real images (Unsplash, etc.)\" and \"Expert-designed scenarios.\" While Unsplash generally has permissive licenses, the inclusion of human subjects in tasks like \"Emotion Change\" or \"Virtual Try-On\" (Section: e000, Multi-Image Tasks) necessitates a clear statement on how consent was obtained or how the dataset complies with privacy regulations (e.g., GDPR, CCPA). If the dataset contains identifiable faces, the authors must confirm that appropriate anonymization or consent protocols were followed, or explicitly exclude such images from the public release.

Third, the **Evaluation Methodology** relies on \"MLLM-as-judge\" using proprietary models like Gemini 3 Pro and GPT-5.1 (Section: e000, Evaluation Pipeline). This introduces a potential safety blind spot. If the judging models have strict safety filters, they may penalize or refuse to evaluate outputs that are actually safe but push the boundaries of the filter, or conversely, fail to detect subtle safety violations that the judge model overlooks. The authors should discuss the limitations of using black-box, safety-filtered models as ground-truth judges and consider how this might bias the benchmark results regarding safety performance.

Finally, the **Reward Modeling** component (\\rmbench) simulates RL optimization scenarios (Section: e000, \\rmbench). The paper should briefly address whether the preference pairs include any \"adversarial\" or \"jailbreak\" attempts to test model robustness, or if the current setup only evaluates benign edits. If the latter, the benchmark may not adequately prepare models for real-world safety challenges.

These issues are primarily writing and disclosure-related (fixable by adding text) but are critical for the ethical deployment of the benchmark.

Action items.

- [**writing**] The manuscript introduces a unified benchmark for image editing and reward modeling. From a safety and ethics perspective, the paper raises several concerns that require clarification before acceptance. First, the Impact Statement (Section: supp:impact) is critically inadequate. The authors state, \"no specific societal consequences highlighted,\" which is a significant oversight given the nature of the tasks evaluated. The benchmark explicitly includes \"Emotion Change\" and \"Object Interactio

Scientific Evidence

Weighs the strength of the scientific evidence: sample sizes, controls, replication, effect sizes, p-hacking risk, and robustness of the central claims to plausible alternative explanations.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_scientific_evidence.md`

The paper presents a substantial benchmark with 2,388 instances and 2,251 preference pairs, which is a significant contribution to the field. However, the scientific evidence supporting the central claims regarding model performance and the validity of the evaluation protocol requires strengthening.

The primary concern is the reliance on MLLM-as-judge for the entire evaluation pipeline (Section 3.3, Appendix A). While the authors claim high correlation with human preferences, the manuscript does not provide standard statistical measures of judge reliability, such as inter-annotator agreement (e.g., Krippendorff's alpha) or variance across different judge models. Without these metrics, it is difficult to rule out systematic biases or hallucinations in the automated scoring, which could invalidate the comparative rankings of the 29 image editing models.

Furthermore, the claim that native multimodal LLMs significantly outperform explicit reward models (Abstract, Section 5.2) is based on point estimates (e.g., 0.7183 vs 0.5587 in Table 1). The paper lacks statistical significance testing (e.g., paired t-tests, bootstrap confidence intervals, or effect sizes) to demonstrate that these differences are not due to random variance. Given the high stakes of these comparisons for the field, providing p-values or confidence intervals is essential.

Finally, the human validation study mentioned in the Appendix (180 instances) is critical for establishing the ground truth of the benchmark. However, the description is sparse. The review requires details on the number of human annotators per instance, the specific rubric they followed, and the exact statistical correlation reported in Figure User_Study. Without this transparency, the claim that the benchmark aligns with human judgment remains an assertion rather than a demonstrated fact.

Action items.

- [**science**] The evaluation relies entirely on MLLM-as-judge (e.g., Gemini 3 Pro, GPT-5.1) without reporting inter-annotator agreement (Krippendorff's alpha) or variance across multiple judge models. Section 3.3 and Appendix A need statistical validation of the judge's reliability to rule out systematic bias or hallucination in scoring.
- [**science**] The claim that 'Native multimodal LLMs outperform explicit reward models' (Abstract, Section 5.2) lacks statistical significance testing (e.g., paired t-tests or bootstrap confidence intervals) on the reported score differences (e.g., 0.7183 vs 0.5587). Without p-values or effect sizes, the robustness of this conclusion is unclear.
- [**science**] The 'Human Evaluation' described in the Appendix (180 instances) is used to validate the MLLM judge but lacks details on the human annotation protocol (e.g., number of annotators per instance, specific rubric used by humans, and the exact correlation metric reported in Figure User_Study). This limits the ability to assess the ground truth validity.

Statistical Analysis

Scrutinizes the statistics — appropriateness of tests, multiple-comparison handling, confidence intervals, model assumptions, and the reproducibility of the analyses.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_statistical_analysis.md`

The statistical rigor of the benchmark construction and evaluation requires clarification to support the paper's claims. While the dataset size (2,388 instances) is substantial, the analysis of results relies heavily on point estimates without measures of uncertainty.

First, the main results tables (e.g., Table 1 in the main text, Tables in the Appendix) present average scores (e.g., Nano-Banana Pro: 3.99) without confidence intervals, standard deviations, or standard errors. Given the variance inherent in generative model outputs and the complexity of the tasks, a single mean value is insufficient to determine if the observed gap between proprietary (3.99) and open-source (2.69) models is statistically significant or driven by outliers. The authors should report the standard error of the mean (SEM) or 95% confidence intervals for all aggregated scores.

Second, the evaluation pipeline aggregates three dimensions (Instruction Awareness, Visual Consistency, Visual Quality) into a single score using a "weighted geometric mean" (Appendix, Evaluation Metrics). The specific weights assigned to each dimension are not explicitly defined in the text. Furthermore, the choice of a geometric mean over an arithmetic mean needs justification, particularly regarding how it penalizes models that perform well in one dimension but poorly in another. Without this transparency, the reproducibility of the ranking is compromised.

Third, the study evaluates 29 models across 36 task categories. This constitutes a massive multiple comparisons scenario. The current analysis identifies "best" models per category without addressing the increased risk of Type I errors (false positives). The authors should clarify if any statistical correction for multiple testing was applied or if the results should be interpreted strictly as descriptive statistics rather than inferential claims of superiority.

Finally, regarding the human annotation stage for \rmbench, the text states that five experts verified pairs for "unanimous agreement." However, no quantitative measure of inter-annotator agreement (such as Cohen's Kappa or Fleiss' Kappa) is reported. For a benchmark of this scale (2,251 pairs), reporting IAA metrics is essential to validate the reliability of the ground truth labels. The absence of these metrics makes it difficult to assess the noise level in the reward modeling dataset.

Action items.

- [**science**] Report confidence intervals or standard errors for all mean scores in Tables 1-5. The current presentation of single-point averages (e.g., 3.99 vs 2.69) lacks statistical context to assess significance or variance across the 2,388 instances.
- [**science**] Clarify the statistical aggregation method for the 'weighted geometric mean' mentioned in Appendix. Explicitly state the weights used for Instruction Awareness, Visual Consistency, and Visual Quality, and justify why a geometric mean is preferred over an arithmetic mean for these specific metrics.
- [**science**] Address the multiple comparisons problem. With 36 distinct task categories and 29 models evaluated, the probability of false positives in identifying 'best' models is high. Specify if any correction (e.g., Bonferroni, FDR) was applied or if the analysis is purely descriptive.
- [**science**] Define the inter-annotator agreement (IAA) metrics for the human evaluation phase. The text mentions 'unanimous agreement' for 2,251 pairs but does not report Cohen's Kappa or Fleiss' Kappa scores to quantify the reliability of the human judgments used to construct the benchmark.

Writing Quality

Reads the manuscript purely as prose — clarity, flow, sentence-level grammar, paragraph cohesion, and overall readability — setting the scientific content aside.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_writing_quality.md`

The manuscript presents a comprehensive benchmark for image editing and reward modeling, but the writing quality requires minor revisions to address specific typographical errors and potential terminology confusion that could distract readers.

First, in the Appendix section (e001), there is a clear typographical error in the label for the system prompts regarding multi-image tasks. The text reads "Instruction Following System Prompts for **Mutli**-Image Tasks". This should be corrected to "**Multi**-Image Tasks" to maintain professional standards.

Second, the text in Section e001 references a figure label as `Fig: Visual_Text_Editing_cn`. In LaTeX, labels containing spaces are generally discouraged and can lead to compilation warnings or broken cross-references depending on the package configuration. It is recommended to rename this label to `Fig:Visual_Text_Editing_cn` (removing the space) to ensure robust compilation and referencing.

Third, in the same section (e001), the text lists "Casual" reasoning tasks alongside Temporal, Math, and Chemical reasoning. In the context of "World Knowledge Reasoning," the intended term is almost certainly "**Causal**" reasoning (dealing with cause-and-effect relationships), rather than "Casual" (which implies relaxed or informal). This is a significant semantic distinction that must be corrected to avoid confusion regarding the benchmark's capabilities.

Finally, while the flow of the introduction and methodology is generally clear, the transition between the description of the benchmark construction and the evaluation pipeline in Section `\bench` could be slightly smoothed. The current text jumps abruptly from "Three strategies" to "Three dimensions." A brief connecting sentence explaining how the construction strategies inform the evaluation dimensions would improve the logical cohesion of the narrative.

These issues are minor and fixable through careful proofreading and text editing, but they should be addressed before final publication to ensure the paper's clarity and professionalism.

Action items.

- [**writing**] In Section 'e001', the phrase 'Mutli-Image Tasks' in the promptbox label contains a typo ('Mutli' instead of 'Multi'). Correct this to ensure professional presentation.
- [**writing**] In Section 'e001', the text references 'Figures~\ref{Fig:ADD}–\ref{Fig: Visual_Text_Editing_cn}'. The label 'Fig: Visual_Text_Editing_cn' contains a space which is non-standard in LaTeX and may cause compilation warnings or broken references. Rename to 'Fig:Visual_Text_Editing_cn'.
- [**writing**] In Section 'e001', the text states 'Casual' reasoning tasks (e.g., 'Figures~\ref{Fig: Temporal}–\ref{Fig: Chemical}'). Given the context of 'World Knowledge', the intended term is likely 'Causal' (cause-and-effect), not 'Casual' (relaxed). Verify and correct this terminology.